Determinants of Land Values in Cebu City, Philippines

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Abstract:

Land is a key asset in wealth creation. It is one of the three sources or factors of production. Since 2008, Cebu City have experienced continued real estate boom and urban growth. Growth begets a myriad of problems such as traffic congestion, housing backlog and unemployment. Land values are possibly the most fundamental factor in urban development planning. Understanding the determinants of land value is a basic requirement in policy making and in addressing the problems. This study will examine the 31 determinants of land value such as accessibility to transportation, neighborhood, distance to city center, neighborhood and others.

The data set used were a result of a survey conducted with 52 real estate practitioners, valuers, assessors randomly selected and uses a 5 likert scale questionaires as a tool In analysing the data set, the research uses factor analysis, principal component analysis and multiple regression with SPSS. In arriving at the ranking of the factors in land values, the study assigned as dependent variable land value data obtained from the Bureau of Internal Revenue (BIR) zonal valuation.Based on the analysis made in this research, the priority and ranking can be categorized into mobility (accessibility to public transportation), livability (open space and parks, neighborhood, recreational facilities, environmental quality), economic (accessibility to employment, rental income), government regulation and taxation (zoning and government assessment and charges), and ownership. Findings of this research could be helpful in urban planning policy and decision-making processes. In particular urban planners, investors, land owners, brokers, valuers could use this model to evaluate the current situation and helps in determining residential land prices.

Keywords: land value; land use sustainable development; real estate; regression

Introduction

Land is a key asset in wealth creation. It is one of the three sources or factors of production. Since 2008, Cebu City have experienced continued real estate boom and urban growth. Growth begets a myriad of problems such as traffic congestion, housing backlog and unemployment. Land values are possibly the most fundamental factor in urban development planning. Understanding the determinants of land value is a basic requirement in policy making and in addressing the urban problems. The real estate is vital for a country's economy as urban land value is determinative in both urban planning and real estate activities in economies of today's world. Knowing the factors affecting the land values is an important advantage in identifying the future of urban development and anticipating probable changes (Topcu, 2009). Land value is both the determinant and the result of location and investment decisions.

This study discusses the determinants of residential land values in Cebu City. Within this scope, the aim of the study is to determine the variables that have significant relationship and contribution to the residential land values. This study is expected to contribute to further studies in city planning, urban design and real estate. For the government, land values are central to understanding property prices and assessments, and the economic impact of land-use policies and taxes levied on property. For private individuals and corporations, land value is one of the basic factors in the location decisions taken by residential and business firms.

Area of Study. Cebu City is the capital city of the province of Cebu and the second urban center of the Philippines. It is the gateway of trade and commerce for the Visayas and Mindanao. It is the economic, trading and educational center of the central and southern Philippines, besides developing as a distribution relay hub and a tourist area (PIDS, 2004) According to the 2015 census, it has a population of 922,611 and area of 326.10 square kilometers and population density of 2,900/sq.km. Population growth of 1.73% for 1999-2000, and 2/3 of population is in 15% ot the total land area. Cebu City has 76.3 % of its land covered under the NIPAS or National Integrated Protected Areas System. Geographically, it is constrained by the sea in the east and mountain and hills in the west.

The Cebu City's Zoning Map (Figure 1) shows that the coastal area is facing the Mactan Strait and predominantly industrial. Commercial areas follow inward, then followed by urban residential areas. As the slope goes higher and higher, the areas are rural residential, followed the Sudlon National Park, as well as other open areas and parks. Pockets of institutional areas are scattered throughout the narrow coastal flatland where the combine of commercial and industrial, as well as urban residential zones can be found.

In this context, 15 vicinities of Cebu City were selected from various barangays with random sampling method. Eight of these vicinities are from the Southern portion and the other seven vicinities comes from Northern portion of the city. The main criterions used in the selection of the vicinities include vcinities that are mostly residential areas and the land values are different despite the house types are similar.

Review of Literatures

There are number of theoretical and empirical literatures on urban land value and its determinants. Previous literatures discusses land value in relation to urban planning, transportation and urban development. Classical insight from Von Thunen, Ricardo and Alonso posited that land values ultimately stem from differences in the quality of locations, ameneties and land use, that is the economic rent of the agricultural land of their time. In their study, Dr. Bolen et al., emphasized that urban policy and planning may be improved by a better understanding of the determinants of urban spatial structure, while (Dr.Bolen et al.). Richardson et al., concluded in their study that Central Business Districts (CBD) did not have statistically significant influence on the spatial distributions of property values, however proximity to seaside is quiete important.(Richardson et al., 1989). Furthermore, the study of Grether and Methowzki found out that the effects of the neighborhood variables appear to be robust.

Gwamma et al emphasizes the importance of utility, interest and willingness to pay (WTP) for attributes as very crucial in any real property decision. Albouy and Enrich argues that land values are central to understanding property prices and assessments, and the economic impact of land-use policies and taxes levied on property. In his paper, Topcu argued that the valuation of any land value, the function of the land location, externalities of the land and the accessibility are the most important factors.

This study examined the determinants of land values and tried to explain the determinants of land values, including the contributions of of accessibility to public transportation, local amenities, access to employment and other variables but expanded the context to include other variables such as legal requirements, government requirements, environment and its relation to land-use regulations and urban form.

Method

Participants

The study focuses on residential land values in Cebu City. The sample is limited to residential properties because residential properties (owner-occupied and investment) represent the largest listings in Cebu City. Furthermore, the market for residential properties is often the most active submarket where sufficient information can be readily garnered for the type of analysis required.

The data for this study were derived from primary and secondary sources. The primary data were obtained through survey from 60 real estate valuers and practitioners, assessor and real estate professionals were contacted during the field survey. The following secondary sources formed the main base for extracting data for the dependent and independent variables: Zonal value from the BIR and Land use map of Cebu City.

A total of 60 survey-questionnaires were distributed through random sampling to respondents in Cebu City, out of which 52 were returned representing about 86 % of the respondents. Out of the returned questionnaires, a total of 51 valid responses were used in the analysis of data. Data (31 by 51 data matrix)relating to the major determinants of land values in the residential neighbourhoods of the City were generated, with the aid of a five-point Likert scale and well-structured questionnaires administered on real estate practitioners and appraisers. This is essentially to test the extent of agreement or disagreement, while the order ranges from 1 as Strongly Disagree, 2 as Disagree, 3 as Undecided, 4 as Agree and 5 as Strongly Agree.

Materials and procedure

The 31 identified major determinants of land values in the residential neighbourhoods of Cebu City include the following namely accessibility to employment, security, environmental quality, view of amenities, city view, level of owner occupation, open spaces and parks, government assessment and charges, capital gains tax, infrastructure availability, electricity, road, sewage disposal, recreational facilities, travel time to city centre, distance to hospital, distance to market, travel time to school, access to main road, access to public transportation, zoning regulations, building types to be built in a residential layout/estate, land/building size ratio, land title, certificate of occupancy, deed of assignment, irrevocable power of attorney, rent, and rental income.

Data were analyzed and evaluated by means of statistical computing package – Statistical Package for Social Sciences (SPSS). In particular, the following statistical techniques were employed namely Factor Analysis (Principal Component Analysis) and multiple regression. Factor analysis is a technique used to reduce a large number of variables into fewer number of factors. This technique extracts maximum common variance from the variables and put them into a common score. Eigenvalues, also called characteristic roots, shows variance explained by that particular factor out of the total variance. From the commonality column, we can know how much variance is explained by the first factor out of the total variance. Then the study employed multiple regression in analyzing the highly loaded and representative factors to come up with the ranking of land value determinants.

Data Analysis And Discussions

Factor analysis and the principal component techniques were applied in the analysis of the independent variables and the relationships between factors influencing residential land values in Cebu City. The 31 determinant was analyzed was done through the application of principal component analysis aimed at making each factor independent of each other. The mechanism of varimax rotation of factor loading based on minimum eigen value of 1.0 was employed.

The results of the principal component analysis displayed in Table 2 shows that eleven (11) factors were extracted. This reduced the number of variables to be looked at in determining land values in Cebu City from the thirty one (31) initially proposed to eleven. In Table 3, loadings in rotated component matrix that are equal or greater than 0.50 are considered to be high.

The result of the principal component analysis showed that 11 eigen values equal to or greater than 1.0 were extracted. Following from the results of Table 4, the 11 factors generated normalized cumulative variance explanation of 82.647% as shown by the rotated sums of squared loadings in Table 3.

Using factor analysis for data reduction, we can determine factors that have highest component loading. The higher the component loadings, the more important that variable for the component. The high loading component factor with less correlation with other was made the representative. The rotated component matrix helps us determine what the component represents. Thus we can observe in Table 3 that the factors that was loaded maximally and has been made as the representative of component 1 is open space and parks (x1). In component 2 factor that loaded maximaly and representative of the component is zoning regulation (x2), while in component 3 factor that represent is government assessments and charges (x3). In component 4 factor that loaded maximally and represents the component is neighborhood (x4), while in component 5 factor that loaded maximally and representative of the component is distance to city center (x5). In component 6, factor that loaded maximally and represent the component is rental income (x6) and in component 7 factor that only loaded maximally is recreational facilities(x7). In component 8 the factor that represent is the accessibility to public transportation (x8) and in component 9 we have the accessibility to employment(x9). In component 10 the factor that loaded maximally is environmental quality (x10) while in component 11 is represented by level of ownership (x11) as a factor that loaded maximally.

It appears that the components are indicators of livability (open space and parks, neighborhood, recreational facilities, environmental quality), government taxation and regulation (zoning and government assessment and charges), economic (rental income, accessibility to employment), and ownership. It resulted in aglomeration of different variables from location, economics, government and legal requirements and externalities.

Application of Multiple Regression. During the study, a Multiple Regression Analysis was employed to examine the contribution of the 11 independent variables in residential land values and arrived at ranking of the factors, using the following model :

$$\mathbf{Y= \beta_0-\beta 1+\beta 2+\beta 3+\beta 4+\beta 6...\beta 11 + e}$$

The land values randomly selected from the BIR data set as dependent variables, since it is considered as consistent and have undergone a definitive process. The study then assigned values to the 11 independet variables to represent its contribution to the dependent variable, and regression analysis were carried. The result is as follows:

Table 4 clearly shows that the contributions of every factor to the land value. The coefficient of -8550.296517 indicates that for every decrease on land value the output will increase by 27.50 . The ranking of factors based on its contribution land values is accessibility to public transportation,recreational facilities, open spaces and parks, environmental quality, level of ownership, government assessment and charges, neighborhood, rental income, distance to city center,zoning and accessibility to employment.

In the output, we can see that the predictor variables of x2, x3,x4,x5,x6,x7,x8 and x9 are significant because both of their P-values are 0.000-0.005.

Findings

From previous researches, the main factors influencing residential values were physical characteristics, location characteristics, and temporal characteristics. This research however has been able to identify 11 variables influencing land values in Cebu City Philippines that can be categorized in economic, livability, location and legal. These made significant contributions to the factors influencing land values in the study areas.

In table 5, it appears that the ranking of factors based on their contributions to the residential land values were mobility (accessibility to public transportation), livability (open space and parks, neighborhood, recreational facilities, environmental quality), economic (accessibility to employment, rental income), government regulation and taxation (zoning and government assessment and charges), and ownership.

Conclusion:

This study has been able to rank the various determinants of residential land values and the strength of the different factors contributing to land value variations in Cebu City.

Based on the findings made in this research, the priority and ranking can be categorized into economic (accessibility to employment, rental income), government regulation and taxation (zoning and government assessment and charges), livability (open space and parks, neighborhood, recreational facilities, environmental quality, and ownership. These major factors need to be given an special consideration in policy formulation and planning the city. It will guide real estate valuers, practitioners, planners and policy makers in making sure that land in the city are put into their highest and best use.

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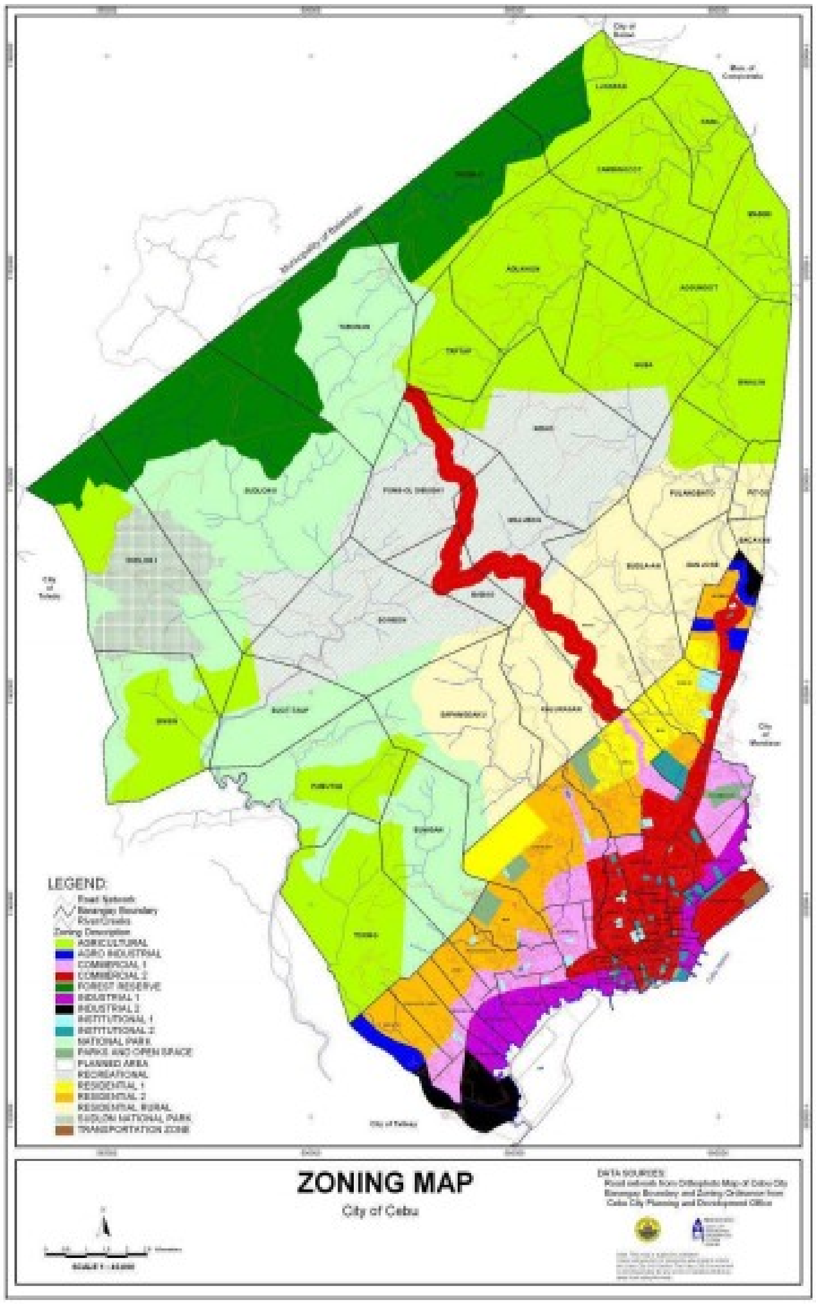
Appendix

Table 1. BIR Zonal Values, 2007

Barangay	Vicinity	Zonal Value/sq.m.
Calamba	Del Rosario-N. Bacalso	8,200.00
Punta Princesa	F. Llamas St	6,700.00
Labangon	Tres de Abril – Katipunan	5,800.00
Apas	Juan Luna Mahiga Bridge	10,000.00
Banilad	Maria Luisa EST Park	13,000.00
Lahug	Junction Gorordo	10,000.00
Mabolo	J. Luna Avenue	10,000.00
Mabolo	Hipodromo-Mandaue	15,800.00
Zapatera	Jakosalem - San Jose Street	14,000.00
Lahug	Salinas Padget Compound	10,000.00
Talamban	Sunny Hills Subdivision	8,000.00
Quiot Pardo	San Carlos Heights Subd	6,100.00
Guadalupe	A. Abellana	6,400.00
Sambag I	Urgello Road	5,600.00
Poblacion Pardo	Abellanosa	4,300.00

Table 2. Rotated Component Matrix[a]

Figure 1: Cebu City Zoning Map



Source: Cebu City Planning Office

	Component										
	1	2	3	4	5	6	7	8	9	10	11
Access	.158	.154	.038	-.248	-.127	-.363	-.116	.312	.514	.351	-.110
Security	.172	.018	-.032	.054	.745	-.095	-.420	.065	.220	-.043	.184
Environmental	.013	-.014	-.036	.028	-.033	.149	.000	-.089	-.065	.852	.041
Amenities	.095	.809	-.062	-.143	.066	-.197	-.090	.239	.147	-.125	.197
View	-.083	.339	-.165	-.025	.724	.054	.322	.056	.094	-.056	-.060
Owner	.044	.093	.016	-.080	.131	-.153	-.102	-.008	.058	-.004	.830
Openspace	.503	.482	-.076	.153	-.043	-.069	.317	.233	.160	.068	.154
Neighborhood	.069	.211	-.044	.683	.130	-.139	.350	.265	-.255	.147	.175
Seaside	.031	-.115	.098	-.030	.153	-.043	.907	-.055	-.023	.032	-.133
Govt charges	.286	.100	-.305	.518	.007	.101	.256	.038	.323	-.008	-.398
CGT	.016	.013	.135	.089	.032	.120	.000	-.049	.897	-.065	.073
Infra facility	.047	.005	.084	.664	.208	-.151	-.034	.260	.082	.444	-.204
Electricity	.180	.140	.028	.859	-.220	.078	-.243	-.049	.090	-.125	-.049
Road	.003	.923	.036	.138	.148	-.051	-.130	.070	-.052	.042	-.047
Sewage	.162	.811	.244	.235	.127	.188	-.012	.066	-.096	.133	-.015
Recreational	.309	.533	.042	.018	-.280	-.038	.444	.299	.296	-.029	.207
Citycenter	.081	.074	.128	-.045	.788	-.053	.170	.201	-.241	.111	.085
DHospital	.628	.020	.174	.084	-.219	.311	.330	-.107	-.005	-.038	.346
DMarket	.621	.056	.151	.051	.096	.380	.046	.052	.128	.358	.254
School	.665	-.010	-.009	.167	.333	.307	.023	.390	-.063	-.057	.108
Mainroad	.244	.207	-.045	.103	.072	-.015	-.033	.723	-.054	-.003	-.177
Transportation	-.121	.195	-.048	.096	.212	.237	.015	.827	.065	-.041	.155
Zoning	-.133	.457	.048	.170	.154	-.262	.221	.090	.095	.498	-.134
Bldgtype	.826	.201	.015	.192	-.105	-.088	.061	-.068	-.035	.090	-.094
Landbuilding	.792	.123	-.017	.069	.026	-.053	-.389	-.020	.125	-.036	.057
Land title	.873	-.072	.017	-.088	.153	-.104	.070	.131	-.002	-.121	-.170
Occupancy	.152	.066	.895	-.059	.076	.138	.120	-.035	.113	.001	.155
Deed	.217	.010	.814	-.149	.001	.346	.052	-.208	-.016	.033	-.045
SPA	-.132	.077	.308	.076	-.031	.818	.011	.195	.027	-.017	-.059
Sublease	-.371	.107	.750	.263	-.152	.014	-.067	.155	.098	-.041	-.101
Rental	-.162	.208	-.120	.155	.057	-.728	.077	-.025	-.072	-.130	.187

Extraction Method: Principal Component Analysis.

Rotation Method: Varimax with Kaiser Normalization.

Table 3: **Total Variance Explained**

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	5.911	19.069	19.069	5.911	19.069	19.069	4.215	13.595	13.595
2	3.593	11.590	30.659	3.593	11.590	30.659	3.341	10.778	24.373
3	2.911	9.390	40.048	2.911	9.390	40.048	2.447	7.893	32.266
4	2.406	7.762	47.810	2.406	7.762	47.810	2.376	7.665	39.931
5	2.245	7.243	55.053	2.245	7.243	55.053	2.301	7.422	47.353
6	1.989	6.416	61.469	1.989	6.416	61.469	2.172	7.007	54.360
7	1.572	5.069	66.538	1.572	5.069	66.538	2.075	6.694	61.054
8	1.500	4.838	71.376	1.500	4.838	71.376	2.021	6.519	67.573
9	1.278	4.123	75.499	1.278	4.123	75.499	1.604	5.174	72.747
10	1.164	3.756	79.254	1.164	3.756	79.254	1.573	5.075	77.822
11	1.052	3.393	82.647	1.052	3.393	82.647	1.496	4.825	82.647
12	.921	2.970	85.617						
13	.853	2.751	88.368						
14	.667	2.152	90.521						
15	.623	2.009	92.529						
16	.526	1.698	94.227						
17	.438	1.412	95.639						
18	.337	1.088	96.727						
19	.249	.804	97.531						
20	.226	.730	98.261						
21	.179	.578	98.839						
22	.110	.356	99.195						
23	.088	.285	99.480						
24	.073	.237	99.717						
25	.031	.101	99.818						
26	.025	.082	99.900						
27	.016	.052	99.952						
28	.009	.028	99.980						
29	.006	.019	99.999						
30	.000	.001	100.000						
31	1.001E-013	1.003E-013	100.000						

Extraction Method: Principal Component Analysis.

Table 4. Coefficients and Significance

	Coefficients	*P-value*
Intercept	-8550.296517	0.001240108
x1	-0.232277359	0.340237716
x2	8.92890491	0.000135451
x3	3.743355456	0.005798919
x4	4.354577867	0.000405649
x5	7.846195041	0.000347218
x6	6.95930096	0.001085239
x7	-6.131085336	0.000371692
x8	-10.06123342	0.000529036
x9	11.44916878	0.001103071
x10	0.623142257	0.322315933
x11	0.029515289	0.976868777

Table 5. Ranking of High Loading Factors

1	Accessibility to public transportation
2	Recreational facilities
3	Open spaces and parks
4	Environmental quality
5	Level of ownership
6	Government assessment and charges
7	Neighborhood
8	Rental income
9	Distance to city center
10	Zoning regulation
11	Accessibility to employment